Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Plane Shapes: Their Types, Properties, Similarities and Differences

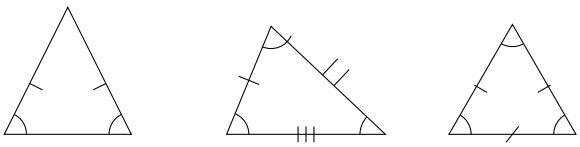
Plane shapes are shapes that lie flat on a surface, so they are also called two-dimensional shapes (2-D shapes). The plane shapes you will meet include the triangle, the square, the rectangle, the parallelogram, the trapezium, the circle and other polygons. A polygon is any closed shape that has three or more straight sides. Let us look at each family of plane shapes, its properties, and how the shapes are similar to or different from one another.

Look at the three triangles in fig-01: the isosceles, the scalene and the equilateral. Their names tell you how their sides compare.

- An isosceles triangle has two adjacent sides equal (the two slanted sides carry a single tick each). Its base angles are equal, and it has one line of symmetry.
- A scalene triangle has no side equal to another — each side carries a different number of ticks (one, two, three). No angle is equal to another, and it has no line of symmetry.
- An equilateral triangle has all three sides equal (a single tick on each side). All three angles are equal, each measuring $60\,^{\circ}$, and it has three lines of symmetry.

So the key differences come down to how many sides are equal: an equilateral has three equal sides, an isosceles has two, and a scalene has none.

Visual



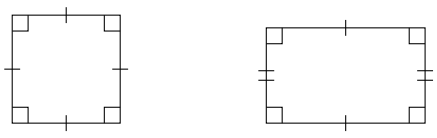
Isosceles Scalene Equilateral

Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene: one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick on each of the three sides, an arc at each vertex.

Now compare the square and the rectangle in fig-02. Each has a right-angle mark at every corner, so both have all angles equal, each measuring $90\,^{\circ}$, and both have two pairs of parallel sides. The difference lies in the sides.

- A square has all four sides equal (a single tick on every side). It has four lines of symmetry, and its diagonals bisect each other at a right angle.
- A rectangle has only two opposite sides equal — the two longer sides carry a single tick each and the two shorter sides a double tick each. It has two lines of symmetry, and its diagonals bisect each other but not at a right angle.

Visual



Squares Rectangles

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

Finally, compare the parallelogram and the trapezium in fig-03.

- A parallelogram has two pairs of opposite sides equal and parallel (its two horizontal sides carry a single arrow and its two slanted sides a double arrow). Its opposite angles are equal, it has no line of symmetry, and its diagonals divide it into two equal triangles.
- A trapezium has only one pair of parallel sides (the two parallel sides carry a single arrow each, and one diagonal is drawn in). Its opposite angles are not equal, it has no line of symmetry, and its diagonal divides it into two unequal triangles.

The great difference between them is the number of parallel sides: a parallelogram has two pairs, a trapezium has only one pair.

Visual



Parallelogram Trapezium

Parallelogram: a single arrow on each of the two horizontal sides, a double arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

Polygons come in many forms, and each is named by its number of straight sides. The table below lists the common polygons, their number of sides, and how many triangles each can be divided into by drawing diagonals from one vertex to all the non-adjacent vertices.

Name of polygon	**Number of sides**	**Number of triangles $(n-2)$**
Triangle	3	1
Quadrilateral	4	2
Pentagon	5	3
Hexagon	6	4
Heptagon	7	5
Octagon	8	6
Nonagon	9	7
Decagon	10	8
Hendecagon (or undecagon)	11	9
Dodecagon	12	10

The rule that produces the third column is simple: take the number of sides n and subtract 2. A hexagon has 6 sides, so it divides into $6-2=4$ triangles; a pentagon has 5 sides, so it divides into $5-2=3$ triangles.

Worked example — triangles in a polygon

How many triangles can a heptagon be divided into if you draw diagonals from one vertex to all the non-adjacent vertices?

Solution. A heptagon has 7 sides. Use the rule: number of triangles $= n-2=7-2=5$. So a heptagon divides into 5 triangles, which matches the table.

Practice

1. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all the non-adjacent vertices?
2. Name the triangle whose three sides are all equal, and state the size of each of its angles.
3. Which of the square and the rectangle has four lines of symmetry?
4. State one difference between a parallelogram and a trapezium.
5. How many sides does a dodecagon have, and into how many triangles can it be divided?

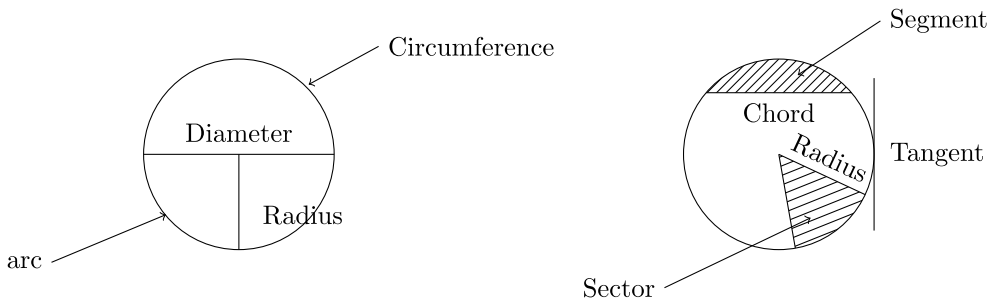
Section 2: Parts of a Circle

A circle is a plane shape with no corners: every point on its boundary is the same distance from its centre. The curved line that forms the round boundary of the circle is its **circumference**. fig-04 shows two labelled circles so you can see each part sitting on a real circle.

- The **centre** is the point in the middle of the circle.
- The **radius** (plural *radii*) is any straight line joining the centre to the circumference. In the second circle of fig-04 a radius is drawn from the centre out to the circumference, and the right-hand bounding line of the sector below is also a radius.
- A **chord** is any straight line joining two points on the circumference. The second circle in fig-04 has a chord drawn near the top.
- The **diameter** is any chord that passes through the centre of the circle. In the first circle of fig-04 the diameter is drawn straight across through the centre. Because it goes through the centre, it is the longest chord a circle can have, and it is twice as long as the radius.
- An **arc** is a curved part of the circumference. In the first circle of fig-04 a part of the circumference is marked and named the arc.

The figure also shows three more parts:
- A **segment** is the region of the circle cut off by a chord — the shaded part above the chord in the second circle.
- A **sector** is the region bounded by two radii and the arc between them — the shaded part below in the second circle, whose two straight sides are both radii.
- A **tangent** is a straight line that touches the circle at exactly one point, as shown touching the right side of the second circle.

Visual



Two labelled circles. The first has the diameter drawn across it and a radius from its centre to the circumference, with the arc and the circumference labelled. The second has a chord near the top with the segment above it shaded, a shaded sector below with its right-hand bounding line labelled Radius, and a tangent touching it on the right.

Practice

1. Name the straight line that joins the centre of a circle to its circumference.
2. What do we call a chord that passes through the centre of the circle?
3. A straight line drawn across a circle joining two points on its circumference is called a _____.
4. The curved part of the circumference between two points on a circle is called an _____.
5. If a circle has a radius of 5 cm, how long is its diameter?
6. Which part of a circle is the region bounded by two radii and the arc between them? Which part is the region cut off by a chord?
7. A straight line that touches a circle at exactly one point is called a _____.

Section 3: Finding the Perimeter of Plane Shapes

The perimeter of a shape is the distance all the way round it. To find it you simply add the lengths of every side. For a straight-sided shape, this means adding each side once. For a circle, the perimeter has a special name — the circumference — and you find it using the radius (or diameter) and the number π (pi), which is about $\frac{22}{7}$ or 3.14.

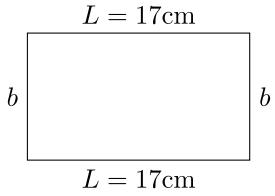
Because some shapes have equal sides, you can often write a short formula instead of adding each side separately. Work through the examples below to see how each shape's perimeter is found.

Example 1 — a square. A square has four equal sides. In fig-05 each side is marked L and measures 15cm. Its perimeter is four lengths added together:
$$L + L + L + L = 4L$$
$$= 15 + 15 + 15 + 15 = 4(15) = 60\text{cm}.$$

Example 2 — a rectangle. A rectangle has two long sides L and two short sides b. In fig-06, $L = 10$cm and $b = 7$cm. Adding all four sides:
$$P = L + b + L + b = 2L + 2b = 2(L + b)$$
$$= 2(10 + 7) = 2(17) = 34\text{cm}.$$

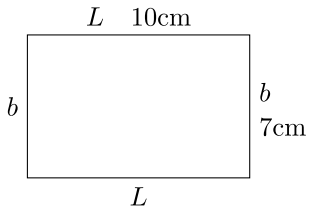
Example 3 — a missing side. Sometimes you are given the perimeter and one side and must find the other. Suppose a rectangle has perimeter 60cm and length 17cm (fig-07). Write the perimeter formula and solve for b:
$$P = 2L + 2b$$
$$60 = 2(17) + 2b$$
$$60 = 34 + 2b$$
Collect like terms:
$$60 - 34 = 2b$$
$$2b = 26$$
Divide both sides by 2:
$$b = 13\text{cm}.$$

Visual



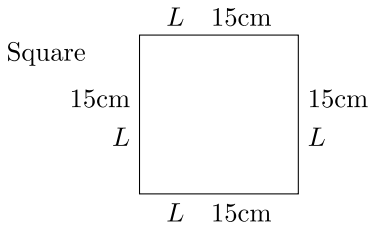
A rectangle with both long sides marked $L = 17$cm and both short sides marked b.

Visual



A rectangle with both long sides marked L, the top one also 10cm, and both short sides marked b, the right-hand one also 7cm.

Visual

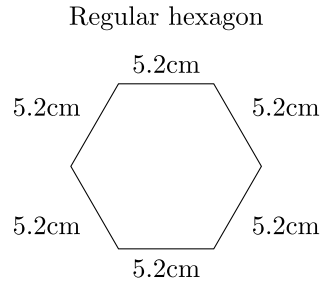


A square, named *Square* at its left, with all four sides marked twice — L and 15cm.

Example 4 — a trapezium. A trapezium has four sides, usually of different lengths. In fig-08 they are $a = 11$cm, $b = 9$cm, $c = 5$cm and $d = 4$cm. The perimeter is all four added:
$$P = a + b + c + d$$
$$= 11 + 9 + 5 + 4 = 29\text{cm}.$$

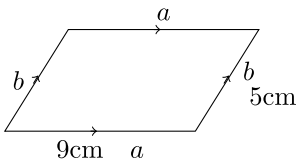
Example 5 — a parallelogram. A parallelogram has opposite sides equal: two of length a and two of length b (fig-09, where $a = 9\text{cm}$ and $b = 5\text{cm}$). So
$$P = a + a + b + b = 2a + 2b = 2(a + b)$$
$$= 2(9 + 5) = 2(14) = 28\text{cm}.$$
Example 6 — a regular hexagon. A regular hexagon has six equal sides. In fig-10 each side is 5.2cm, so
$$P = 6 \times 5.2 = 31.2\text{cm}.$$

Visual



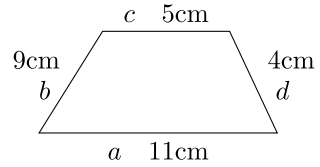
A regular hexagon, named above the drawing, with each of its six sides marked 5.2cm.

Visual



A parallelogram with a single arrow on each of its four sides: the two horizontal ones both lettered a, the lower also measured 9cm, and the two slanted ones both lettered b, the right-hand one also measured 5cm.

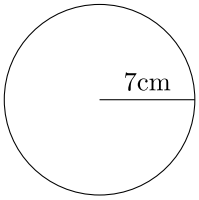
Visual



A trapezium with each side lettered and measured: the top c, 5cm; the left b, 9cm; the right d, 4cm; the base a, 11cm.

Example 7 — a circle. The perimeter of a circle is its circumference, found with either formula:
$$C = 2\pi r \quad \text{or} \quad C = \pi d.$$
In fig-11 the radius is 7cm, using $\pi = \frac{22}{7}$:
$$C = 2 \times \tfrac{22}{7} \times 7 = 2 \times 22 = 44\text{cm}.$$
Using πd instead: since $d = 2r = 2 \times 7 = 14\text{cm}$,
$$C = \tfrac{22}{7} \times 14 = 22 \times 2 = 44\text{cm}.$$
Example 8 — a rope wound round a cylinder. One turn round the cylinder is one circumference. With radius 25cm and $\pi = 3.14$,
$$C = 2\pi r = 2 \times 3.14 \times 25 = 157\text{cm}.$$
Wound 50 times, the rope's total length is
$$157 \times 50 = 7850\text{cm}.$$
Example 9 — the radius from the circumference. If the circumference is 440cm, find r ($\pi = \frac{22}{7}$):
$$440 = 2\pi r = 2 \times \tfrac{22}{7} \times r = \tfrac{44r}{7}.$$
Cross-multiply:
$$440 \times 7 = 44r \quad \Rightarrow \quad 3080 = 44r.$$
Divide both sides by 44:
$$r = \tfrac{3080}{44} = 70\text{cm}.$$

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked 7cm.

Example 10 — a quarter circle. The shape in fig-14 is made of two straight sides of 4.9cm each and a curved part that is one quarter of a full circle. Its perimeter is the two straight sides plus that curved part.

First find the curved part. It is $\frac{1}{4}$ of a full circumference:

$$\frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9$$

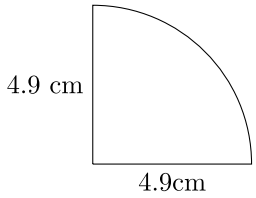
$$= \frac{1}{4} \times \frac{44}{7} \times 4.9 = \frac{11}{7} \times \frac{49}{10}$$

$$= \frac{11 \times 7}{10} = \frac{77}{10} = 7.7\text{cm}.$$

Now add the straight sides and the curved part:

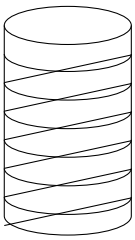
$$P = 4.9 + 4.9 + 7.7 = 17.5\text{cm}.$$

Visual



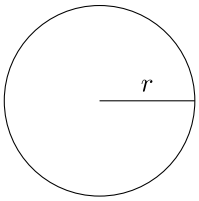
A quarter circle: an upright side and a base meeting at a right angle, each marked 4.9cm, with an arc closing them from the top of the upright to the end of the base.

Visual



An upright cylinder with a rope wound round it, drawn as a series of turns spiralling down the body from the top face to the base.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked r.

Practice

1. Find the perimeter of a square with sides of 8cm.
2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
3. What is the perimeter of a regular pentagon with sides of 5cm?
4. Find the perimeter of a parallelogram whose base is 6cm and whose sides are 5cm.
5. The circumference of a circle is 22cm. Find its radius ($\pi = \frac{22}{7}$).

6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle.
7. Find the perimeter of a regular hexagon with sides of 5cm.
8. A trapezium has sides 11cm, 9cm, 5cm and 4cm. Find its perimeter.

Check your work

Attempt every question before you read this page.

Plane Shapes: Their Types, Properties, Similarities and Differences

1. 6 triangles

   An octagon has 8 sides. Number of triangles $= n - 2 = 8 - 2 = 6$.
2. It is an equilateral triangle, and each of its three angles measures 60°.
3. The square has four lines of symmetry; the rectangle has two.
4. A parallelogram has two pairs of parallel sides, while a trapezium has only one pair of parallel sides.
5. A dodecagon has 12 sides and divides into 10 triangles.

   Number of triangles $= n - 2 = 12 - 2 = 10$.

Parts of a Circle

1. The radius.
2. The diameter.
3. chord.
4. arc.
5. 10 cm.

   The diameter is twice the radius, so 2 × 5 cm = 10 cm.
6. The region bounded by two radii and the arc between them is a sector; the region cut off by a chord is a segment.
7. tangent.

Finding the Perimeter of Plane Shapes

1. 32cm

   A square has four equal sides, so $P = 4 \times 8 = 32\text{cm}$.
2. 22cm

   $P = 2L + 2b = 2(7) + 2(4) = 14 + 8 = 22\text{cm}$.
3. 25cm

   A regular pentagon has five equal sides, so $P = 5 \times 5 = 25\text{cm}$.
4. 22cm

   Opposite sides are equal, so $P = 2a + 2b = 2(6) + 2(5) = 12 + 10 = 22\text{cm}$.
5. 3.5cm

   $C = 2\pi r$, so $22 = 2 \times \frac{22}{7} \times r = \frac{44r}{7}$. Cross-multiply: $22 \times 7 = 44r$, so $154 = 44r$, giving $r = \frac{154}{44} = 3.5\text{cm}$.
6. 6d

   Length $= 2d$ and breadth $= d$. Perimeter $= 2(\text{length} + \text{breadth}) = 2(2d + d) = 2(3d) = 6d$.
7. 30cm

   A regular hexagon has six equal sides, so $P = 6 \times 5 = 30\text{cm}$.
8. 29cm

   $P = 11 + 9 + 5 + 4 = 29\text{cm}$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-390e45d9	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____